

David Santiago Puentes

Full-Stack Developer · Applied AI · Electronics and IoT

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Portfolio

Profile

Full-stack developer with more than two years of professional experience and an electronic-engineering background. I work with web applications, APIs, databases, automation, firmware, and IoT. I use agents, subagents, skills, MCP, and token budgets with human validation. Demonstrable technical details are backed by public projects; employment experience is presented in accordance with confidentiality obligations.

Professional Experience

Full-Stack Developer · Private technology-solutions company

Aug 2025 — 2026 · Colombia

- Developed and maintained full-stack web applications as part of a technology team.
- Worked with interfaces, APIs, relational databases, authentication, authorization, and secure-development practices.
- Contributed to requirements discovery, implementation, deployment, support, and continuous improvement of business software.

Research and Development Developer · CEINTECCI

2023 — 2025 · Colombia

- Conducted applied work in scientific software, electronics, sensing, wireless communications, and IoT.
- Developed reproducible processes for data acquisition, cleaning, analysis, and visualization.
- Contributed to innovation work with a granted patent and science outreach.

Selected Public Projects

SmartSense — LoRa and industrial IoT monitoring

PT100/MAX31865 sensors, ESP32 nodes, LoRa communications, a gateway, and a local historical dashboard. Team thesis project with a public repository and evidence.

Sensor Dashboard — web IoT platform

MicroPython and LoRa → Flask API → PostgreSQL → Next.js, TypeScript, and Tailwind dashboard. Public end-to-end project.

Project H — room and asset security

BLE nodes, a custom binary protocol, an ESP32 master with a physical alarm, and a Firebase-backed web application. Documented prototype.

Technical Skills

Backend: PHP, Python, Node.js, REST APIs, Flask, relational modeling

Frontend: JavaScript, TypeScript, React, Next.js, Astro, Tailwind CSS

Data: MySQL/MariaDB, PostgreSQL, SQLite, pandas, NumPy, SciPy

Electronics and IoT: ESP32, C/C++, MicroPython, LoRa, BLE, PT100/MAX31865, KiCad

Delivery and security: Git/GitHub, Docker, Linux, automated tests, authentication, authorization, CSRF, and request limiting

AI: agents and subagents, skills, MCP, token budgets, context control, and human review

Education and Certifications

Electronic Engineering — Universidad ECCI (in progress) · Technologist in Industrial Electronics — Universidad ECCI (2023) · English B2 — Universidad ECCI (2025)

AI Application Development · React.js with TypeScript · TypeScript · Terraform · Cybersecurity and Privacy for Businesses — Platzi (2026)
Scientific Computing with Python · Responsive Web Design — freeCodeCamp (2025)